EXCITATION-DEPENDENT IDENTIFIABILITY OF LATENT HIGHER-ORDER STRUCTURE FROM EDGE FLOWS

Weiping Yan

University of Minnesota, Twin Cities
Minneapolis, MN, USA

ABSTRACT

Which triangles of a network carry higher-order interactions? Edge flows reveal the latent filled-triangle set $\mathcal{S}$ only through their curl, and we show that *what* is identifiable is governed by the excitation covariance $\boldsymbol{\Gamma}_{\mathcal{S}}$ of the triangle signals, through three regimes. For positive-diagonal $\boldsymbol{\Gamma}_{\mathcal{S}}$ (known or unknown), the weighted support is identifiable at *any* rank deficiency of $\boldsymbol{B}_2$, because the lifted signatures $\{\boldsymbol{b}_\tau \boldsymbol{b}_\tau^\top\}$ are always linearly independent. For unrestricted PSD $\boldsymbol{\Gamma}_{\mathcal{S}}$, the population curl covariance determines $\boldsymbol{M} = \boldsymbol{\Sigma}_z - \sigma_n^2 \boldsymbol{I}$ exactly but pins the support only up to $\{\mathcal{S} : \operatorname{im} \boldsymbol{M} \subseteq \operatorname{im} \boldsymbol{U}_{\mathcal{S}}\}$ (the achievable set is $\mathcal{C}_{\mathcal{S}} = \{\boldsymbol{M} \succeq 0 : \operatorname{im} \boldsymbol{M} \subseteq \operatorname{im} \boldsymbol{U}_{\mathcal{S}}\}$); a strictly positive-*diagonal* but correlated $\boldsymbol{\Gamma}_{\mathcal{S}}$ already defeats identification (a rank-one witness on K_4). At the projector excitation $\boldsymbol{\Gamma}_{\mathcal{S}} = \sigma_c^2 (\boldsymbol{B}_{2,\mathcal{S}}^\top \boldsymbol{B}_{2,\mathcal{S}})^+$ the covariance collapses to $\sigma_n^2 \boldsymbol{I} + \sigma_c^2 \boldsymbol{P}_{\mathrm{im}}$ exactly, so equal-image supports are indistinguishable at every SNR and sample size (noise level known throughout). This projector is exactly the Hodge-smoothness prior used in prior topology learning [1]. In the diagonal regime we give a lifted-covariance nonnegative least-squares estimator that is consistent with a fully explicit $O(1/N)$ failure bound, validated on K_4–K_8 against subspace and greedy baselines, including an excitation interpolation under which equal-image distinguishability provably vanishes. On real data, the same curl statistic *localizes* genuine unplanted rotational structure — tropical cyclones in ERA5 winds — against a same-field vorticity reference and an external IBTrACS archive. Code, data, and a 52-test guardrail suite (the quoted constants checked against Monte-Carlo) are released.

Index Terms— Topological signal processing, simplicial complexes, identifiability, Hodge decomposition, covariance estimation, edge flows.

1. INTRODUCTION

Edge flows — currency exchange, traffic, ranking comparisons — carry higher-order structure on the triangles of a simplicial complex [2, 3, 4]. Topological signal processing builds filters and detectors on the Hodge Laplacian [5, 6, 7, 8, 9], and a recurring primitive is that the set $\mathcal{S}$ of "filled" triangles is unknown. One line *estimates* $\mathcal{S}$ algorithmically (MAP under a Hodge-smoothness prior [1], greedy learning [10], sparse cell-complex augmentation [11]); a second-order line models the edge covariance (PCA/total-variation on the sample covariance [2, §VII], ML fitting of a Hodge-structured simplicial Gaussian model [12]); a third *detects* signals in a *known* Hodge subspace [13]. Each fixes a model and gives an algorithm; to our knowledge none characterizes, as a function of the excitation class, *when* the filled-triangle structure is identifiable from the edge covariance — the gap this paper fills: *what part of the latent structure is identifiable, and when?*

The answer is not a property of the geometry alone: it depends on the *excitation*. We model triangle signals as $\boldsymbol{y}_{\mathcal{S}} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Gamma}_{\mathcal{S}})$ with PSD excitation covariance $\boldsymbol{\Gamma}_{\mathcal{S}}$ and show that the identifiable object moves with the regime of $\boldsymbol{\Gamma}_{\mathcal{S}}$ (Theorem 1): from the full weighted support (positive diagonal, any rank deficiency), to only the achievable-covariance set $\mathcal{C}_{\mathcal{S}}$ — so the support merely up to $\{\mathcal{S} : \operatorname{im} \boldsymbol{M} \subseteq \operatorname{im} \boldsymbol{U}_{\mathcal{S}}\}$ — under unrestricted PSD excitation, to provable indistinguishability of equal-image supports at the projector excitation. The last case is not a corner curiosity: the Hodge-smoothness prior of [1], $\boldsymbol{y}_{\mathcal{S}} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{L}_2^+)$, $\boldsymbol{L}_2 = \boldsymbol{B}_2^\top \boldsymbol{B}_2$, *is* this projector excitation — equal-image supports then have identical marginalized covariance likelihood, so any choice between them comes from the prior, not the data. We thus correct a folklore "rank obstruction" and delimit what covariance- and image-level methods [2, 12, 13] can resolve. Our tools are classical — detection theory and Fano arguments [14, 15], in the lineage of support-recovery and sparse-covariance limits [16, 17, 18, 19] — applied through a reduction (curl annihilation, after HodgeRank's curl-as-inconsistency reading [20]) that makes them bite. Contributions:

1. *Theory*: the three excitation regimes (Thm. 1), with global analytic separation constants for the isotropic regime — equal-image binary supports are separated by at least 9 (attained by adding a tetrahedron's fourth face) and equal-cardinality supports by 16 (swaps), proved for *every* clique complex via a Johnson-graph eigenvalue argument — and Chernoff exponents, for fixed graph and support as $\rho_2 \to 0$, whose worst case equals one isolated-triangle detection (Thm. 2).
2. *Achievability*: a lifted-covariance NNLS estimator for the diagonal regime, consistent with a fully explicit $O(1/N)$ failure bound in which the constants are derived and numerically spot-checked (Thm. 3); Monte-Carlo validation on K_4–K_8 with random supports against oracle-aided subspace and greedy baselines, and an excitation-interpolation experiment in which equal-image distinguishability collapses as predicted (Fig. 1).
3. *Real data, carefully positioned*: curl-based *vortex localization* — not topology recovery — of unplanted tropical-cyclone circulation in ERA5 winds, validated against a same-field vorticity reference and the independent IBTrACS archive, with a classical baseline and moving-block bootstrap confidence intervals (Fig. 2).

2. MODEL AND CURL REDUCTION

Fix a graph $(\mathcal{V}, \mathcal{E})$ with node–edge incidence $\boldsymbol{B}_1$ and, for a candidate triangle set $\mathcal{T}$ (3-cliques), edge–triangle incidence $\boldsymbol{B}_2 \in \mathbb{R}^{|\mathcal{E}| \times p}$;

Code/data: `github.com/appleweiping/topo-flow-limits`.

$\boldsymbol{B}_1\boldsymbol{B}_2 = \mathbf{0}$, and each column $\boldsymbol{b}_\tau$ has three ±1 entries, so $\boldsymbol{G} = \boldsymbol{B}_2^\top\boldsymbol{B}_2$ has $G_{\tau\tau} = 3$ and $G_{\sigma\tau} \in \{0, \pm1\}$ (nonzero iff the triangles share an edge). We observe N i.i.d. snapshots

$$\boldsymbol{f}_t = \boldsymbol{B}_1^\top\boldsymbol{a}_t + \boldsymbol{B}_{2,\mathcal{S}}\,\boldsymbol{y}_t + \boldsymbol{h}_t + \boldsymbol{n}_t, \qquad t = 1,\dots,N, \quad (1)$$

with gradient nuisance $\boldsymbol{a}_t \sim \mathcal{N}(\mathbf{0}, \sigma_g^2\boldsymbol{I})$, harmonic nuisance $\boldsymbol{h}_t$, noise $\boldsymbol{n}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2\boldsymbol{I})$, and triangle excitation

$$\boldsymbol{y}_t \sim \mathcal{N}(\mathbf{0},\ \boldsymbol{\Gamma}_\mathcal{S}), \qquad \boldsymbol{\Gamma}_\mathcal{S} \succeq 0, \quad (2)$$

where $\boldsymbol{\Gamma}_\mathcal{S} \succeq 0$ is the excitation covariance; making it explicit is the point of this paper (prior generative models are special cases of it, §1). We take the harmonic nuisance to be *candidate-orthogonal*, $\boldsymbol{h}_t \in \ker\boldsymbol{B}_2^\top$ (orthogonal to the curl space of the *full* candidate complex); the gradient part is annihilated unconditionally since $\boldsymbol{B}_2^\top\boldsymbol{B}_1^\top = \mathbf{0}$. Then curl projection removes both nuisances exactly [20]: in an orthonormal basis $\boldsymbol{Q}$ of $\operatorname{im}\boldsymbol{B}_2$ ($r = \operatorname{rank}\boldsymbol{B}_2$, $\boldsymbol{B}_2$ the full candidate incidence), the sufficient statistic $\boldsymbol{z}_t = \boldsymbol{Q}^\top\boldsymbol{f}_t$ is Gaussian with

$$\boldsymbol{\Sigma}_z(\mathcal{S}, \boldsymbol{\Gamma}_\mathcal{S}) = \sigma_n^2\boldsymbol{I}_r + \boldsymbol{U}_\mathcal{S}\boldsymbol{\Gamma}_\mathcal{S}\boldsymbol{U}_\mathcal{S}^\top, \qquad \boldsymbol{U} = \boldsymbol{Q}^\top\boldsymbol{B}_2, \quad (3)$$

with $\|\boldsymbol{u}_\tau\|^2 = 3$ and $\boldsymbol{u}_\sigma^\top\boldsymbol{u}_\tau = G_{\sigma\tau}$. Identifiability of $\mathcal{S}$ from (3) is the object of study; for the isolated-triangle scalar test this reduces to the classical two-variance problem with curl-SNR $\rho = 3\sigma_c^2/\sigma_n^2$ [14] (supplement §S1–S3 develops that line: detection thresholds, Fano converses, decorrelated detectors, plug-in estimation).

Definition 1 (Observational equivalence and identifiability)**.** *All statements here are about the* population $\boldsymbol{\Sigma}_z$ *(finite-N estimation is §4). Two pairs $(\mathcal{S}, \boldsymbol{\Gamma}_\mathcal{S})$ and $(\mathcal{S}', \boldsymbol{\Gamma}_{\mathcal{S}'})$ are* observationally equivalent *if $\boldsymbol{\Sigma}_z(\mathcal{S}, \boldsymbol{\Gamma}_\mathcal{S}) = \boldsymbol{\Sigma}_z(\mathcal{S}', \boldsymbol{\Gamma}_{\mathcal{S}'})$; a functional of $(\mathcal{S}, \boldsymbol{\Gamma}_\mathcal{S})$ is* identifiable *on an excitation class $\mathcal{G}$ if it is constant on each equivalence class with excitations in $\mathcal{G}$. The* realized range *is $\mathcal{R}(\mathcal{S}, \boldsymbol{\Gamma}_\mathcal{S}) = \operatorname{im}(\boldsymbol{\Sigma}_z - \sigma_n^2\boldsymbol{I}) = \operatorname{im}(\boldsymbol{U}_\mathcal{S}\boldsymbol{\Gamma}_\mathcal{S}^{1/2}) \subseteq \operatorname{im}\boldsymbol{U}_\mathcal{S}$, always identifiable since it is a function of $\boldsymbol{\Sigma}_z$ alone. If $\boldsymbol{\Gamma}_\mathcal{S} \succ 0$ then $\mathcal{R} = \operatorname{im}\boldsymbol{U}_\mathcal{S}$; when $\boldsymbol{B}_{2,\mathcal{S}}$ has full column rank (so $\boldsymbol{U}_\mathcal{S}$ is injective) the converse holds as well.*

3. EXCITATION-DEPENDENT IDENTIFIABILITY

Lemma 1 (Lifted spark)**.** *For any candidate set of distinct 3-cliques of a simple graph, the matrices $\{\boldsymbol{b}_\tau\boldsymbol{b}_\tau^\top\}_{\tau\in\mathcal{T}}$ are linearly independent.*

Proof. A pair of distinct edges lies in at most one common triangle, so for $e \neq e'$ both in τ, the (e, e') entry of $\sum_\sigma w_\sigma\boldsymbol{b}_\sigma\boldsymbol{b}_\sigma^\top$ equals $\pm w_\tau$: every coefficient is read off a distinct matrix entry. ☐

Theorem 1 (Three excitation regimes)**.** *Let σ_n^2 be known and write $\boldsymbol{M} = \boldsymbol{M}(\mathcal{S}, \boldsymbol{\Gamma}) = \boldsymbol{U}_\mathcal{S}\boldsymbol{\Gamma}\boldsymbol{U}_\mathcal{S}^\top = \boldsymbol{\Sigma}_z - \sigma_n^2\boldsymbol{I}$ in (3). The regimes below are overlapping constraints on $\boldsymbol{\Gamma}_\mathcal{S}$, not a partition (e.g. $\sigma_c^2\boldsymbol{I}$ is both diagonal (a) and a special full-rank PSD (b)).* (a) Positive diagonal. *If $\boldsymbol{\Gamma}_\mathcal{S} = \operatorname{diag}(\gamma_\tau)_{\tau\in\mathcal{S}}$ with $\gamma_\tau > 0$ (known or unknown), then $\boldsymbol{M} = \sum_{\tau\in\mathcal{S}}\gamma_\tau\boldsymbol{u}_\tau\boldsymbol{u}_\tau^\top$ and, by Lemma 1, the map $(\mathcal{S}, \boldsymbol{\Gamma}_\mathcal{S}) \mapsto \boldsymbol{\Sigma}_z$ is injective: the weighted support is identifiable at* any *rank deficiency of $\boldsymbol{B}_2$, including all $\binom{n}{3}$ triangles of K_n.* (b) Arbitrary PSD. *The achievable set $\mathcal{C}_\mathcal{S} = \{\boldsymbol{U}_\mathcal{S}\boldsymbol{\Gamma}\boldsymbol{U}_\mathcal{S}^\top : \boldsymbol{\Gamma} \succeq 0\} = \{\boldsymbol{M} \succeq 0 : \operatorname{im}\boldsymbol{M} \subseteq \operatorname{im}\boldsymbol{U}_\mathcal{S}\}$, so the* population *covariance determines $\boldsymbol{M}$ in full but pins the support only to $\{\mathcal{S} : \operatorname{im}\boldsymbol{M} \subseteq \operatorname{im}\boldsymbol{U}_\mathcal{S}\}$: the realized range $\mathcal{R} = \operatorname{im}\boldsymbol{M} \subseteq \operatorname{im}\boldsymbol{U}_\mathcal{S}$ is always readable, yet the candidate image $\operatorname{im}\boldsymbol{B}_{2,\mathcal{S}}$ is not (a larger-image $\mathcal{S}'$ with a suitable $\boldsymbol{\Gamma}$ gives the same $\boldsymbol{M}$; $\boldsymbol{\Gamma}_\mathcal{S} \succ 0$ does recover $\mathcal{R} = \operatorname{im}\boldsymbol{U}_\mathcal{S}$). The obstruction is* correlation, *not zero variance: on K_4 (all four faces) with $\boldsymbol{c} \in \ker\boldsymbol{U}_\mathcal{S}$ and $\boldsymbol{v} = \boldsymbol{e}_1 - \frac{1}{4}\boldsymbol{c} = (.75, .25, -.25, .25)^\top$, the strictly-positive-diagonal $\boldsymbol{\Gamma}' = \boldsymbol{v}\boldsymbol{v}^\top$ gives $\boldsymbol{U}_\mathcal{S}\boldsymbol{\Gamma}'\boldsymbol{U}_\mathcal{S}^\top = \boldsymbol{u}_1\boldsymbol{u}_1^\top$ — identical to the singleton $\{\tau_1\}$ though the images are 3- vs. 1-dimensional. Regime (a) additionally requires $\boldsymbol{\Gamma}$ diagonal.* (c) Projector excitation. *For $\boldsymbol{\Gamma}_\mathcal{S} = \sigma_c^2(\boldsymbol{B}_{2,\mathcal{S}}^\top\boldsymbol{B}_{2,\mathcal{S}})^+$, $\boldsymbol{M} = \sigma_c^2\boldsymbol{P}_{\operatorname{im}\boldsymbol{U}_\mathcal{S}}$* exactly*; supports with equal curl image induce* identical *snapshot distributions and are indistinguishable at every SNR and every N.*

Proof sketch. (a) Read the weights off the lifted representation (spark). (b) $\subseteq$ is immediate; $\supseteq$: given PSD $\boldsymbol{M}$ with $\operatorname{im}\boldsymbol{M} \subseteq \operatorname{im}\boldsymbol{U}_\mathcal{S}$, take $\boldsymbol{\Gamma} = \boldsymbol{U}_\mathcal{S}^+\boldsymbol{M}(\boldsymbol{U}_\mathcal{S}^+)^\top \succeq 0$. (c) $\boldsymbol{A}(\boldsymbol{A}^\top\boldsymbol{A})^+\boldsymbol{A}^\top = \boldsymbol{P}_{\operatorname{im}\boldsymbol{A}}$ (SVD). On K_n, $\dim\operatorname{im}\boldsymbol{B}_2 = \binom{n-1}{2}$ (a $3/n$ DoF ratio), and equal-image supports exist whenever a tetrahedron is hosted ($\boldsymbol{b}_{012} - \boldsymbol{b}_{013} + \boldsymbol{b}_{023} - \boldsymbol{b}_{123} = \mathbf{0}$). Full proofs and numerical certificates: supplement §S4, `tests/test_excitation.py`. ☐

Remark 1 (Scope)**.** *(i) Deterministic unknown signals are the rank-one degenerate boundary of regime (b): the realized range is a single direction, so not even the image is seen. (ii) On K_n with σ_n* unknown*, regime (a) retains exactly one ambiguous direction, since $\sum_{\tau\in\mathcal{T}}\boldsymbol{u}_\tau\boldsymbol{u}_\tau^\top = n\boldsymbol{I}_r$: a uniform weight shift trades off against the noise floor. (iii) Regime (b) delimits image-level methods — e.g. matched subspace detectors [13] — whose population statistics see only the realized range $\operatorname{im}\boldsymbol{M} \subseteq \operatorname{im}\boldsymbol{B}_{2,\mathcal{S}}$: without excitation structure, no method can do better.*

Theorem 2 (The price in the isotropic class)**.** *Let $\boldsymbol{\Gamma}_\mathcal{S} = \sigma_c^2\boldsymbol{I}$, $\rho_2 = \sigma_c^2/\sigma_n^2$, and write the dimensionless signature Gram $\boldsymbol{D}_\mathcal{S} = \sum_{\tau\in\mathcal{S}}\boldsymbol{u}_\tau\boldsymbol{u}_\tau^\top$ (so $\boldsymbol{M}_\mathcal{S} = \sigma_c^2\boldsymbol{D}_\mathcal{S}$). For* any *clique complex: distinct binary supports with equal curl image satisfy $\|\boldsymbol{D}_\mathcal{S} - \boldsymbol{D}_{\mathcal{S}'}\|_F^2 \geq 9$, with equality exactly for single-face differences (e.g. $\mathcal{S}' = \mathcal{S} \cup \{\text{fourth face}\}$), and ≥ 16 under equal cardinality (tetrahedral swaps). For fixed graph and support pair, as $\rho_2 \to 0$ the optimal-test Chernoff exponents are $C_G = \rho_2^2\|\Delta\boldsymbol{D}\|_F^2/16\,(1 + o(1))$, i.e. $\frac{9}{16}\rho_2^2(1+o(1))$ (subset) and $\rho_2^2(1+o(1))$ (swap) — and $\frac{9}{16}\rho_2^2$* equals *the isolated-triangle detection exponent $C(\rho) = \rho^2/16$ at $\rho = 3\rho_2$: the hardest equal-image decision costs one isolated-triangle detection. Hence* in this scope *(fixed pair, isotropic excitation, leading low-SNR exponent) rank deficiency is free at the exponent level, and only the Fano log-multiplicity of the $4\binom{n}{4}$ tetrad hypotheses ($\asymp \log n$ on K_n) reflects the geometry.*

Proof sketch. Separations: $\|\sum_\tau c_\tau\boldsymbol{b}_\tau\boldsymbol{b}_\tau^\top\|_F^2 = \boldsymbol{c}^\top(9\boldsymbol{I} + \boldsymbol{A})\boldsymbol{c}$ for ternary $\boldsymbol{c}$, where $\boldsymbol{A}$ is the share-an-edge adjacency — an induced subgraph of the Johnson graph $J(n, 3)$, so $\lambda_{\min}(\boldsymbol{A}) \geq -3$ by interlacing and sep $\geq 6\|\boldsymbol{c}\|^2$; enumerate $\|\boldsymbol{c}\|^2 \in \{1, 2\}$. Exponents: Bhattacharyya expansion gives $C_G = \rho_2^2\|\Delta\boldsymbol{D}\|_F^2/16\,(1 + o(1)) = \|\Delta\boldsymbol{M}\|_F^2/(16\sigma_n^4)\,(1 + o(1))$ (the two forms coincide). Numerically checked: swap C_G/ρ_2^2 within 6×10^{-4} of 1 and subset matching the single-triangle exponent to 3×10^{-5} relative (at $\rho_2 = 10^{-4}, 10^{-5}$); separations exhaustive over all 2^{10} supports of K_5. Proofs: supplement §S4. ☐

Proposition 1 (Edge-sharing marginal laws)**.** *Assume $\boldsymbol{B}_2$ full column rank and $\boldsymbol{\Gamma}_\mathcal{S} = \sigma_c^2\boldsymbol{I}$. The GLS/BLUE inversion $\hat{\boldsymbol{y}}_t = \boldsymbol{G}^{-1}\boldsymbol{B}_2^\top\boldsymbol{f}_t$ [21] satisfies, exactly, $\hat{\boldsymbol{y}}_t = \tilde{\boldsymbol{y}}_t + \boldsymbol{e}_t$ with $\boldsymbol{e}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2\boldsymbol{G}^{-1})$: each coordinate is a two-variance test with effective SNR $\rho_\tau^{\text{eff}} = \sigma_c^2/(\sigma_n^2(\boldsymbol{G}^{-1})_{\tau\tau})$ — the classical variance-inflation factor as the price of edge-sharing — and exact recovery admits a correlation-robust union-bound guarantee. The noise coordinates stay correlated ($\sigma_n^2\boldsymbol{G}^{-1}$ is not diagonal), so the joint law does not*

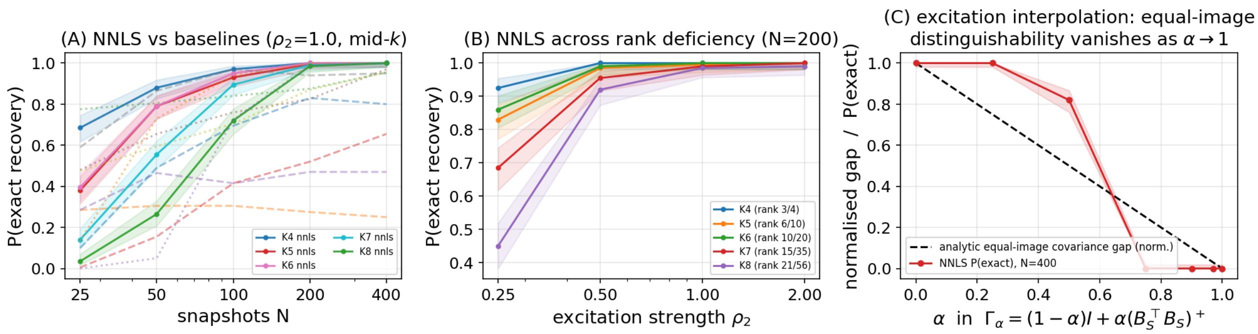

Fig. 1. Second-order achievability. (A) Exact recovery vs. N on K_4–K_8 (random supports, $\rho_2 = 1$): NNLS (solid, 95% Wilson bands) vs. oracle-aided subspace (dashed) and greedy (dotted) baselines. (B) NNLS vs. ρ_2 at $N = 200$ across rank deficiency (legend: rank $\boldsymbol{B}_2/p$). (C) Excitation interpolation on the K_5 tetrad: the analytic equal-image gap (normalized) closes only at $\alpha = 1$ (Thm. 1(c)); thresholded NNLS recovery collapses earlier (zero from $\alpha = 0.75$).

factorize; benchmarks, the naive-detector failure mode, and proofs are in supplement §S1, S3.

Proposition 2 (Detection and joint-recovery scalings)**.** *In the isotropic class, deciding one isolated triangle is the two-variance Gaussian test with variances $3\sigma_n^2$ vs. $9\sigma_c^2 + 3\sigma_n^2$ [14]: attaining error $\le \delta$ at the exponent level requires the curl-SNR scale $\rho \ge \rho^\star(N) = \Theta(\sqrt{\log(1/\delta)/N})$ ($C(\rho) = \rho^2/16 + o(\rho^2)$ [15]). Recovering a uniformly random size-k support among p edge-disjoint candidates with error $\le \epsilon$ requires $N \ge [(1-\epsilon)\log\binom{p}{k} - \log 2]/(k\,\mathrm{KL}_1)$, $\mathrm{KL}_1 = (\rho - \log(1+\rho))/2$ (Fano): joint recovery pays a $\log\binom{p}{k}$ factor. Signal-agnostic and sub-Gaussian variants, the exact edge-disjoint product law, partial edge observation (q^{3k} laws), and plug-in calibration are developed in supplement §S1–S3; these scalings are classical [16] — what is problem-specific is the reduction that produces ρ and the constants* $(3, 9)$.

4. ACHIEVABILITY: LIFTED-COVARIANCE NNLS

In class (a) the theory says the weighted support is identifiable; the matching estimator is non-negative least squares on the lifted dictionary $\boldsymbol{A} = [\mathrm{vec}(\boldsymbol{u}_\tau \boldsymbol{u}_\tau^\top)]_\tau$:

$$\hat{\boldsymbol{w}} = \arg\min_{\boldsymbol{w}\ge 0} \left\| \hat{\boldsymbol{\Sigma}}_z - \sigma_n^2 \boldsymbol{I} - \textstyle\sum_\tau w_\tau \boldsymbol{u}_\tau \boldsymbol{u}_\tau^\top \right\|_F^2, \tag{4}$$

with $\hat{\boldsymbol{\Sigma}}_z = \frac{1}{N}\sum_t \boldsymbol{z}_t \boldsymbol{z}_t^\top$, $\hat{\mathcal{S}} = \{\tau : \hat{w}_\tau > w_{\min}/2\}$, and $w_{\min} = \min_{\tau\in\mathcal{S}} w_\tau$.

Theorem 3 (Consistency and explicit failure bound)**.** *$\hat{\boldsymbol{w}} \to \boldsymbol{w}$ almost surely, and for every N,*

$$\Pr[\hat{\mathcal{S}} \ne \mathcal{S}] \;\le\; \frac{16\left[(\mathrm{tr}\,\boldsymbol{\Sigma}_z)^2 + \|\boldsymbol{\Sigma}_z\|_F^2\right]}{N\,\sigma_{\min}^2(\boldsymbol{A})\,w_{\min}^2}. \tag{5}$$

Proof sketch. Three derived steps, each numerically checked: (i) cone-constrained least squares obeys the deterministic bound $\|\hat{\boldsymbol{w}} - \boldsymbol{w}\|_2 \le 2\|\hat{\boldsymbol{\Sigma}}_z - \boldsymbol{\Sigma}_z\|_F/\sigma_{\min}(\boldsymbol{A})$ — optimality of $\hat{\boldsymbol{w}}$, feasibility of $\boldsymbol{w}$, Cauchy–Schwarz — with $\sigma_{\min}(\boldsymbol{A}) > 0$ by Lemma 1; (ii) for Gaussian snapshots $\mathbb{E}\|\hat{\boldsymbol{\Sigma}}_z - \boldsymbol{\Sigma}_z\|_F^2 = [(\mathrm{tr}\,\boldsymbol{\Sigma}_z)^2 + \|\boldsymbol{\Sigma}_z\|_F^2]/N$ *exactly* (Wishart second moments); (iii) Markov at threshold $w_{\min}/2$. The bound is conservative (Markov) — its value is the explicit $O(1/N)$ rate with computable constants; empirically the transition occurs roughly an order of magnitude earlier (Fig. 1). Proof: supplement §S4. □

Fig. 1 validates the estimator across K_4–K_8 with per-trial *random* supports at several sparsities k, a full (N, ρ_2) grid with $N \in \{25, \ldots, 400\}$ and $\rho_2 \in \{0.25, \ldots, 2\}$, 200 Monte-Carlo trials per cell, and 95% Wilson intervals (`run_second_order.py`; every cell, including the derived bound (5) evaluated at that cell, is in the released JSON). Protocol: the headline threshold is the theorem's $w_{\min}/2$ (planted weights known); an oracle-free largest-gap rule is logged alongside but is *worse* (grid-mean 0.63 vs. 0.69, ahead in 18/260 cells). The full edge-domain pipeline is used; the greedy comparator is a generic covariance-atom heuristic, and both baselines get *oracle* side information (subspace: true k and $\dim\mathrm{im}\,\boldsymbol{B}_{2,\mathcal{S}}$; greedy: true σ_n).

(A) Among these three implementations (not tuned literature methods), NNLS is the only one with exact recovery in *every* cell — incl. maximal rank deficiency (rank $\boldsymbol{B}_2/p = 21/56$ on K_8) — reaching 1 at mid-k by N=400, ρ_2=1 (0.99 at the densest K_8 cell, k=28). The oracle-k subspace baseline can lead at small N but collapses on dependent candidates (0.00 at K_8, k=28): its *population* scores tie there (Remark 1(iii)), and its finite-N tie-breaking rides on sample eigen-anisotropy, a side channel the projector excitation removes entirely. Greedy stalls at sparse supports (0.43 on K_5 at k=2, N=400). (B) Recovery improves monotonically in ρ_2 uniformly across rank deficiency — the empirical face of Theorem 1(a). The bound (5) upper-bounds every cell; being a Markov bound it is conservative by one to three orders of magnitude (the empirical failure decays exponentially, the bound as $1/N$). It is non-vacuous (clipped value < 1) at only 7 of the 260 grid cells — the small-$w_{\min}$, small-N corner, where it falls as low as 0.42 (and to $\sim$0.75 at dedicated test cells) with zero observed failures; elsewhere the $O(1/N)$ rate is the guarantee. (C) Under $\boldsymbol{\Gamma}_\alpha = \rho_2[(1-\alpha)\boldsymbol{I} + \alpha(\boldsymbol{B}_{2,\mathcal{S}}^\top \boldsymbol{B}_{2,\mathcal{S}})^+]$ on the K_5 tetrad, with the same $w_{\min}/2$ rule (the diagonal model is deliberately misspecified for $\alpha > 0$ — that is the point), the analytic equal-image gap shrinks continuously and the empirical NNLS recovery of the specific $\mathcal{S}$ collapses *before* the gap reaches zero — thresholded estimation dies earlier than distinguishability — with the threshold-free endpoint at $\alpha = 1$ on Theorem 1(c): the m=4 equal-image supports of the planted cardinality have identical distributions, so no estimator beats chance $1/m$; the plotted 0 is *specific-$\mathcal{S}$* recovery, itself $\le 1/m$. Identifiability is a property of the excitation, not the geometry alone. (A matrix-free form scaling to $p\sim 10^4$ on GPU and a non-oracle BIC threshold are in supplement §S6.)

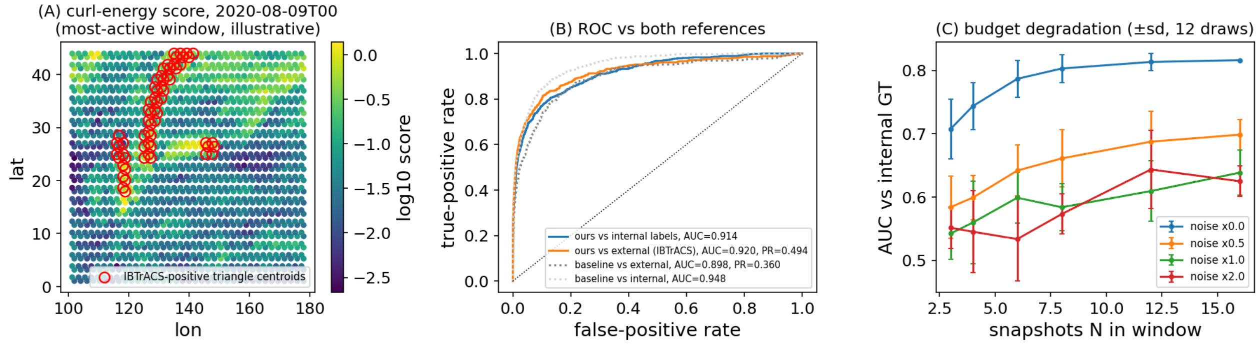

Fig. 2. Curl-based vortex localization on real data. (A) Centered curl-energy scores (vorticity scale) for the single most-active 4-day window (argmax IBTrACS-positive triangles; illustrative only — all metrics pool every window); red circles: *IBTrACS-positive triangle centroids* (not raw fix positions). (B) ROC pooled over all 15 windows: same-field vorticity reference, external IBTrACS labels (with PR-AUC), and the classical coarse-vorticity baseline. (C) Budget degradation (mean $\pm$ sd over 12 subsample draws; uniform centered statistic, $N \geq 3$).

5. REAL DATA: CURL-BASED VORTEX LOCALIZATION

We position the real-data study precisely: it is *vortex localization*, not filled-triangle topology recovery — the mesh below is simply connected, hence full-column-rank with no equal-image confusers, and real weather carries no latent boolean support. What it demonstrates is that the paper's sufficient statistic localizes *genuine, unplanted* rotational structure on a real vector field. Construction: ERA5 10 m winds [22] over the Western North Pacific (Aug–Sep 2020, $0.7°$/6-hourly) are subsampled to a $2.1°$ right-triangulated mesh (836 nodes, 2389 edges, 1554 triangles; #triangles $= |\mathcal{E}| - |\mathcal{V}| + 1$ by Euler, so $\boldsymbol{B}_2$ is full column rank); edge flows are trapezoidal line integrals of the wind in a local equirectangular metric. By Stokes' theorem a triangle's curl is its circulation, so the temporally centered curl-energy score (area^2-normalized to the mean-vorticity scale; no oracle parameters) ranks cyclone-bearing triangles within 4-day windows of 16 snapshots, pooled over the season's 15 windows for evaluation. Storms translate several triangle-widths per 4-day window; centering removes the window-mean flow, so the statistic responds to the time-varying circulation a moving vortex deposits. Validation (Fig. 2B) against a *same-field consistency reference* (finite-difference vorticity on the 3×-finer source grid) gives AUC 0.914 (moving-block bootstrap 95% CI $[0.894, 0.936]$), and against the *external, separately curated* IBTrACS archive [23] (raw: 726 fixes over 13 storms; entering the metric: 414 IBTrACS-positive of 23,310 triangle-windows, prevalence 1.8%) AUC 0.920 $[0.879, 0.951]$, PR-AUC 0.494 $[0.273, 0.666]$, and P@k $=$ 0.483. Against a classical pointwise-vorticity baseline at the same information budget (mesh-node winds), the edge-flow statistic is better on *all three* independent external metrics (baseline: AUC 0.898, PR-AUC 0.360, P@k $=$ 0.389) and slightly lower on the internal reference (0.914 vs. 0.948), which shares the baseline's functional. The decorrelated (GLS) variant of Prop. 1 ranks worse here ($\boldsymbol{G}^{-1}$ amplifies noise on a near-regular mesh) and is reported in the repository. FX consistency checks and planted road-network studies from earlier revisions are retained in the repository and supplement, not claimed as recovery evidence.

Limitations. The i.i.d.-Gaussian snapshot model is an idealization of windowed weather; evaluation is therefore threshold-free ranking rather than support recovery, moving-block *and* storm-cluster bootstrap intervals (which agree) quantify pooling uncertainty and are *exploratory* (one 2020 season), and the internal reference is a same-field consistency check — IBTrACS is the external archive. The AUC is stable across localization radius, wind cutoff, window length, and vorticity threshold (supplement). Nothing here claims a new cyclone detector; the experiment grounds the paper's sufficient statistic on genuinely unplanted structure.

6. CONCLUSION

Identifiability of latent higher-order structure from edge flows is excitation-dependent: fully identifiable under diagonal excitation at any rank deficiency; under unrestricted PSD excitation the population covariance pins down only the achievable-covariance set $\mathcal{C}_{\mathcal{S}}$, so the support is determined merely up to $\{\mathcal{S} : \operatorname{im} \boldsymbol{M} \subseteq \operatorname{im} \boldsymbol{U}_{\mathcal{S}}\}$ (not even the candidate image in general); and provably ambiguous at the projector excitation — with global analytic separation constants, a worst-case exponent equal to one isolated-triangle detection, and a consistent NNLS estimator carrying a fully explicit $O(1/N)$ bound. The three regimes delimit both algorithmic structure learning [1, 10, 11] and subspace detection [13]; on real data the same statistic localizes unplanted cyclone circulation. Open problems: identifiability *between* the diagonal and free-PSD excitation classes (e.g. banded or Toeplitz $\boldsymbol{\Gamma}_{\mathcal{S}}$ — where between reading off every weight and seeing only the image does the transition occur?); sharp first-order sparse thresholds over $\ker \boldsymbol{B}_2$; effective sample sizes N_{eff} under temporal dependence; and tightening the Markov factor in (5) to match the empirical transition.

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Compliance with ethical standards

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